# Sensitivity analysis

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

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WORKBOOK · 1 OF 15

## Recall and choose your starting point

> Record your starting explanation for Sensitivity analysis before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

### Ablation studies

Without opening the old lesson, explain one key idea from Ablation studies. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

### Statistical tests and assumptions

Without opening the old lesson, explain one key idea from Statistical tests and assumptions. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

### Sampling variability

Without opening the old lesson, explain one key idea from Sampling variability. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

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WORKBOOK · 2 OF 15

## Practise with fading help

> Complete the supported Sensitivity analysis trace and record the first step you can explain without help.

Use W2: Trace the worked case for **Sensitivity analysis**. Record the answer once; the lesson and workbook refer to the same attempt.

| Given inputs | Procedure I chose | Middle steps | Result and check |
|---|---|---|---|
| | | | |

**The step I completed without help, and the hint I needed**

WORKBOOK · 3 OF 15

## Individual reasoning check

> Check the proposed Sensitivity analysis explanation against the supplied facts before discussing it.

Use the worked case for **Sensitivity analysis**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

**My explanation before partner discussion**

WORKBOOK · 4 OF 15

## Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

### Independent case: Apply a new case independently

Thresholds .4, .5, .6 give passes8, 9, 7 out of10. Report range and a selection limit.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

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WORKBOOK · 5 OF 15

## W1 Explain the mechanism

> Explain the mechanism. Apply the supplied Sensitivity analysis facts and keep your reasoning.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

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WORKBOOK · 6 OF 15

## W2 Trace the worked case

> Trace the worked case. Apply the supplied Sensitivity analysis facts and keep your reasoning.

Retrieval cutoffs1, 3, 5 yield recall .4, .7, .8 and precision .9, .7, .5. Show the intermediate steps and give the conclusion with its scope.

**My reasoning and evidence**

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## W3 Audit the limit

> Audit the limit. Apply the supplied Sensitivity analysis facts and keep your reasoning.

A report omits this qualification: A stable tested range does not establish stability under every parameter or dataset change. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

**My reasoning and evidence**

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## W4 Apply a new case independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply a new case independently. Apply the supplied Sensitivity analysis facts and keep your reasoning.

Thresholds .4, .5, .6 give passes8, 9, 7 out of10. Report range and a selection limit.

**My reasoning and evidence**

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WORKBOOK · 9 OF 15

## W5 Design a revealing follow-up

> Design a revealing follow-up. Apply the supplied Sensitivity analysis facts and keep your reasoning.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

**My reasoning and evidence**

WORKBOOK · 10 OF 15

## Project checkpoint and reflection

> Connect Sensitivity analysis to Reproducible AI Research Study, identify a limit and choose the next useful test.

Add a labelled **sensitivity analysis evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of **sensitivity analysis**.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

**My project change, supporting evidence and remaining limit**

WORKBOOK · 11 OF 15

## Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

| Evidence | My record |
| --- | --- |
| Worked alone / hint / model | |
| First step I repaired and why | |
| New case checked later; date and result | |

| Evidence | My record |
| --- | --- |
| What I can now explain without notes | |
| One remaining question | |

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

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**WORKBOOK · 12 OF 15**

## Pictorial case practice: Sensitivity analysis

> A decision sensitive to the cutoff. A toy score rule predicts positive when score is at least the chosen cutoff. Explain the labelled step before checking the changed condition.

A decision sensitive to the cutoff. A toy score rule predicts positive when score is at least the chosen cutoff.

Scores are 0.4,0.6,0.8. Cutoff 0.5 marks the last two positive; cutoff 0.7 marks only 0.8 positive.

AI ACADEMY / AI-7/17

## Score decisions under inclusive cutoffs

| | Score 0.4 | Score 0.6 | Score 0.8 |
|---|---|---|---|
| Cutoff 0.5 | No | Yes | Yes |
| Cutoff 0.7 | No | No | Yes |
| Cutoff 0.6 | No | Yes | Yes |

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/17

# Sensitivity analysis

**Start with the supplied facts**

Follow each arrow. Ask which fact allows the next step.



### 1. Hold scores fixed

Keep the same three supplied scores.



### 2. Apply cutoff 0.5

0.6 and 0.8 meet the boundary.



### 3. Apply cutoff 0.7

Only 0.8 meets the new boundary.

**CHANGE ONE THING**

Set the cutoff to exactly 0.6.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Classify 0.4, 0.6 and 0.8 at cutoffs 0.5, 0.7 and exactly 0.6. Which equality rule decides the boundary, and can these unlabelled scores identify the best cutoff?

**My picture-based explanation and changed-case prediction**

### Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning.

Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

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## Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Reproducible AI Research Study record.

Project: Reproducible AI Research Study

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

| Evidence to keep | My record |
| --- | --- |
| This week's input and deciding step | |
| Prediction and actual result (or labelled paper trace) | |
| One change since the previous checkpoint and why | |
| One error or limit and the next test | |

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

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WORKBOOK · 14 OF 15

## Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A decision sensitive to the cutoff. A toy score rule predicts positive when score is at least the chosen cutoff.

### Practice 1: explain the deciding evidence

Which labelled step matches this detail? Only 0.8 meets the new boundary.

1. Hold scores fixed

2. Apply cutoff 0.7

3. Apply cutoff 0.5

**My selected conclusion, deciding fact and explanation**

### Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

### Feedback · compare after trying

After your attempt, compare with the supported explanation: Apply cutoff 0.7 Only 0.8 meets the new boundary. **Sensitivity analysis** checks how conclusions change under plausible choices.

Changing a cutoff changes the decision rule, not the recorded scores or their accuracy.

**Practice 2: explain the deciding evidence**

### Changed case · predict first

Changed-case practice: Set the cutoff to exactly 0.6. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. This one example guarantees correct results for every future **input**.

3. 0.6 and 0.8 are positive because the rule includes equality.

**My selected conclusion, deciding fact and explanation**

### Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

### Feedback · compare after trying

After your attempt, compare with the supported explanation: 0.6 and 0.8 are positive because the rule includes equality. The declared at-least boundary includes the threshold value.

### Repair a missing building block

Review Ablation studies (ai-7/16). Goal: Explain the **mechanism** and terminology of ablation studies. Short practice: Calculate the filter ablation’s two rates and count change.

Which single component is altered, and why does also changing the model prevent isolating the filter?. Return to this week and explain what you changed.

Review Statistical tests and assumptions (ai-7/14). Goal: Explain the **mechanism** and terminology of statistical tests and assumptions. Short practice: Calculate each paired difference and the observed sum.

Count null arrangements with absolute sum at least four, compute p, and explain the decision at threshold 0.05.. Return to this week and explain what you changed.

Review Sampling variability (ai-7/11). Goal: Explain the **mechanism** and terminology of sampling variability. Short practice: Calculate the two four-case rates and their difference.

Explain how changed included cases can affect a fixed model, and why identical deterministic records should not differ on a rerun.. Return to this week and explain what you changed.

### Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

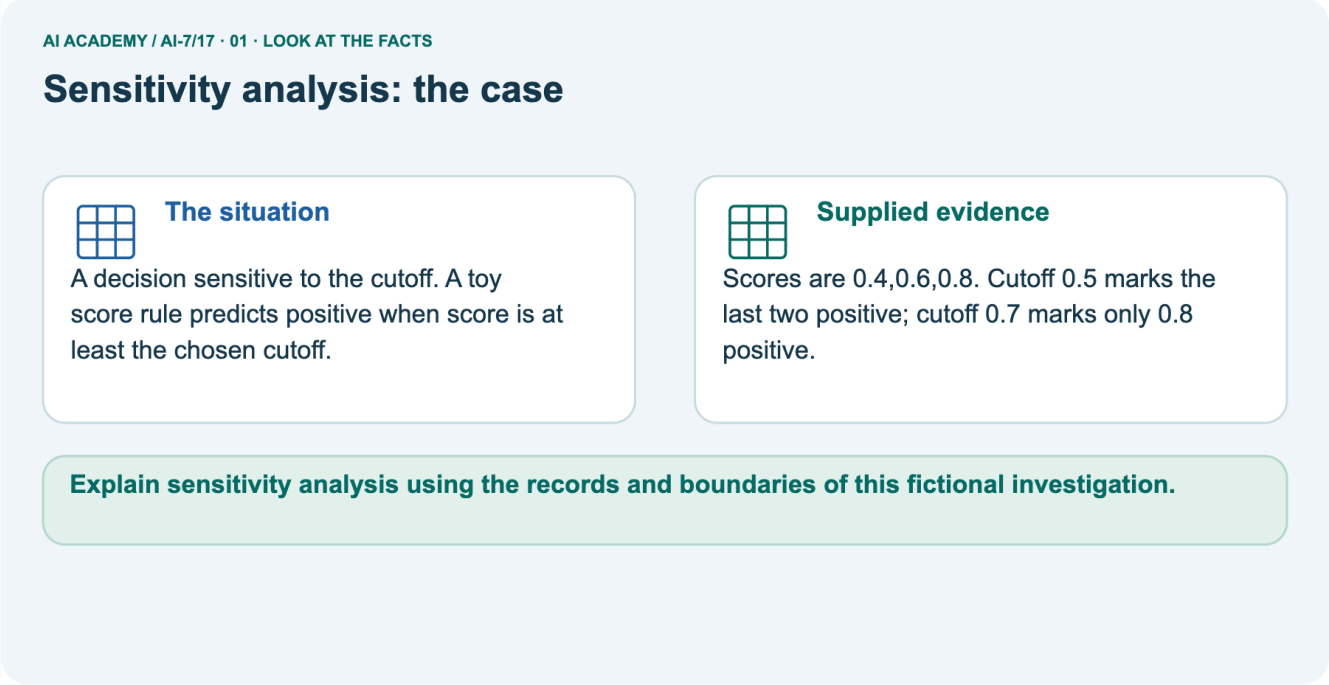
**What I repaired, the evidence I kept and the next test**

WORKBOOK · 15 OF 15

# Visual concept journey

> Explain each Sensitivity analysis picture from its labels, then record a relevant limit.

### 01. Sensitivity analysis: the case




Follow the labelled picture with the matching explanation. Open the vector diagram

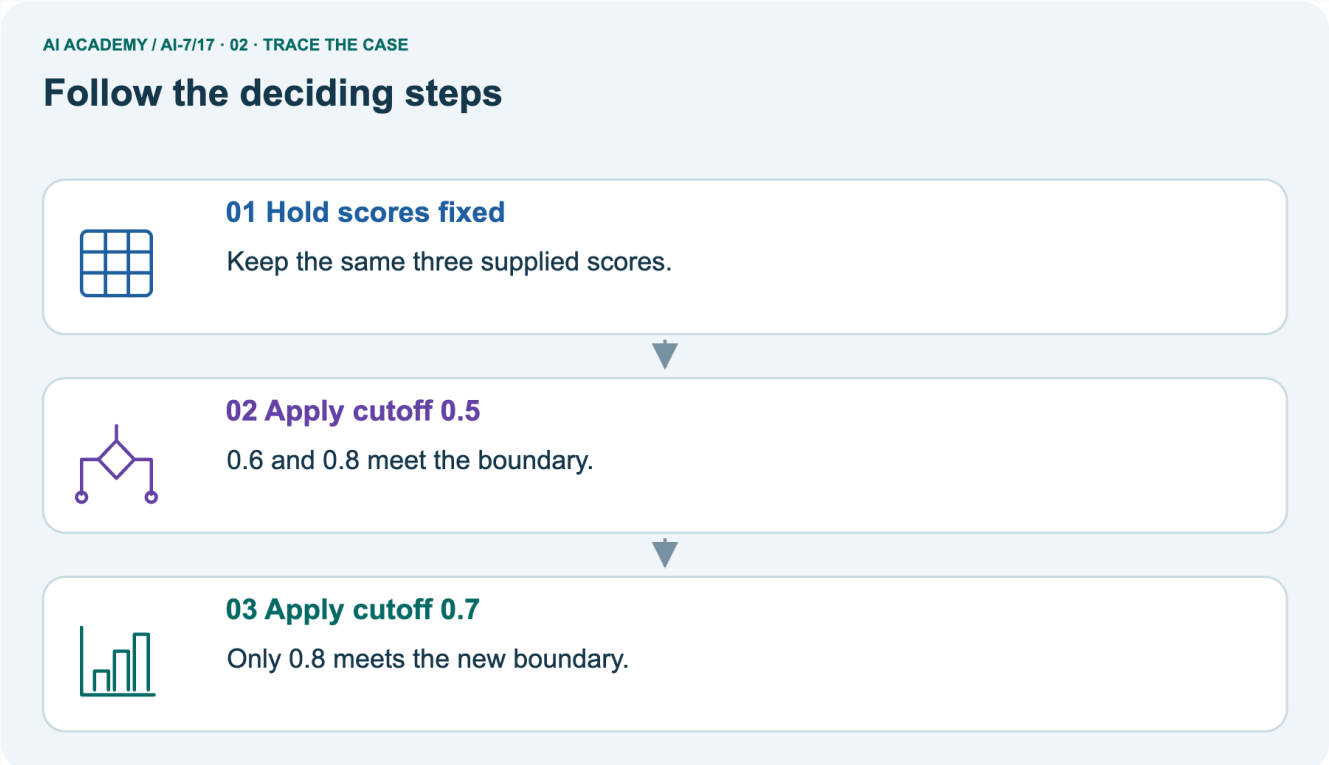
A decision sensitive to the cutoff. A toy score rule predicts positive when score is at least the chosen cutoff. Supplied facts: Scores are 0.4,0.6,0.8.

Cutoff 0.5 marks the last two positive; cutoff 0.7 marks only 0.8 positive.

> Point to the given fact that matters most. Explain why it matters.

**My picture explanation, deciding evidence and one limit**

## 02. Follow the deciding steps



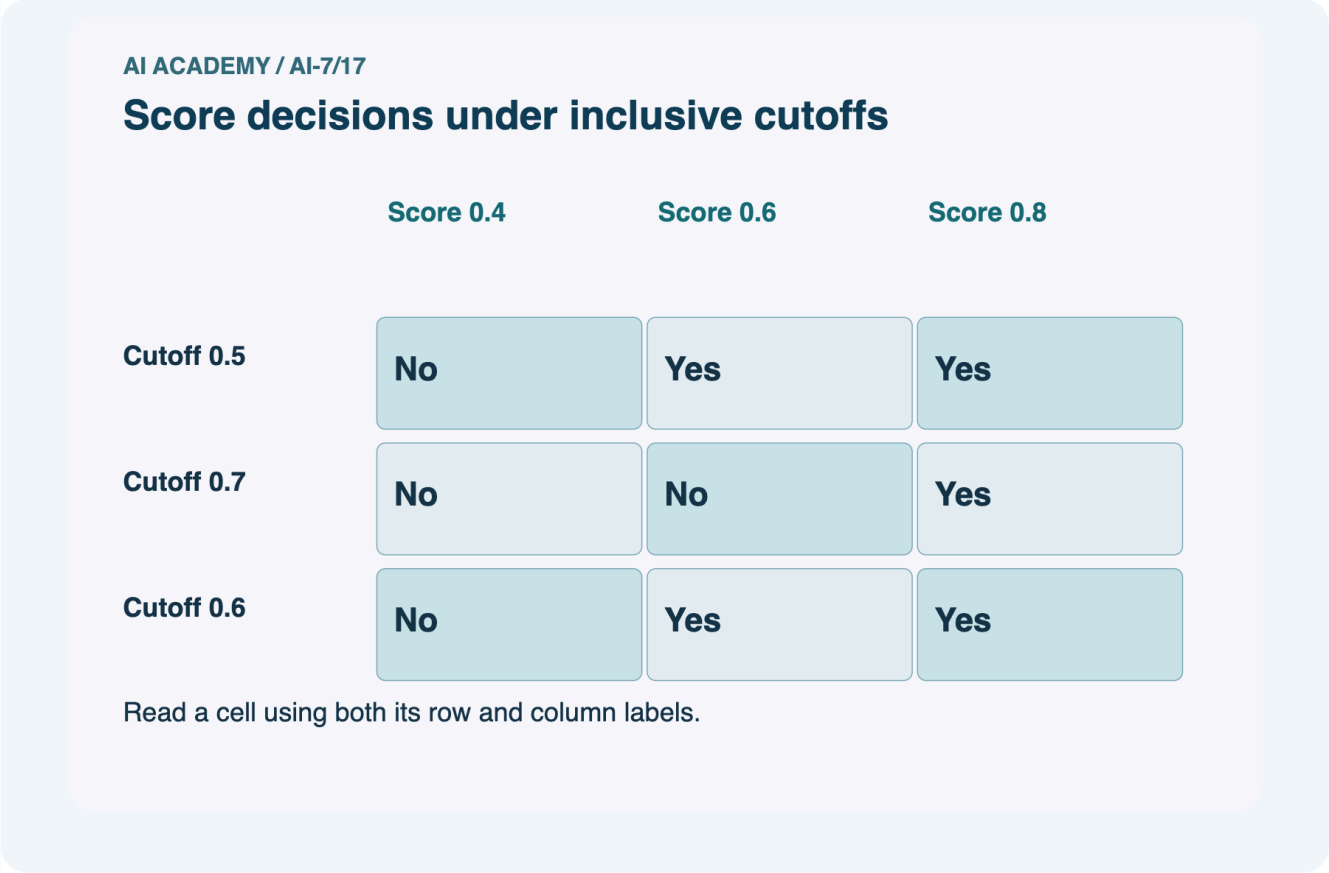

Follow the labelled picture with the matching explanation. Open the vector diagram

Hold scores fixed: Keep the same three supplied scores. → Apply cutoff 0.5: 0.6 and 0.8 meet the boundary. → Apply cutoff 0.7: Only 0.8 meets the new boundary.

Point to each numbered stage and explain which recorded **input** it uses.

**My picture explanation, deciding evidence and one limit**

### 03. Score decisions under inclusive cutoffs



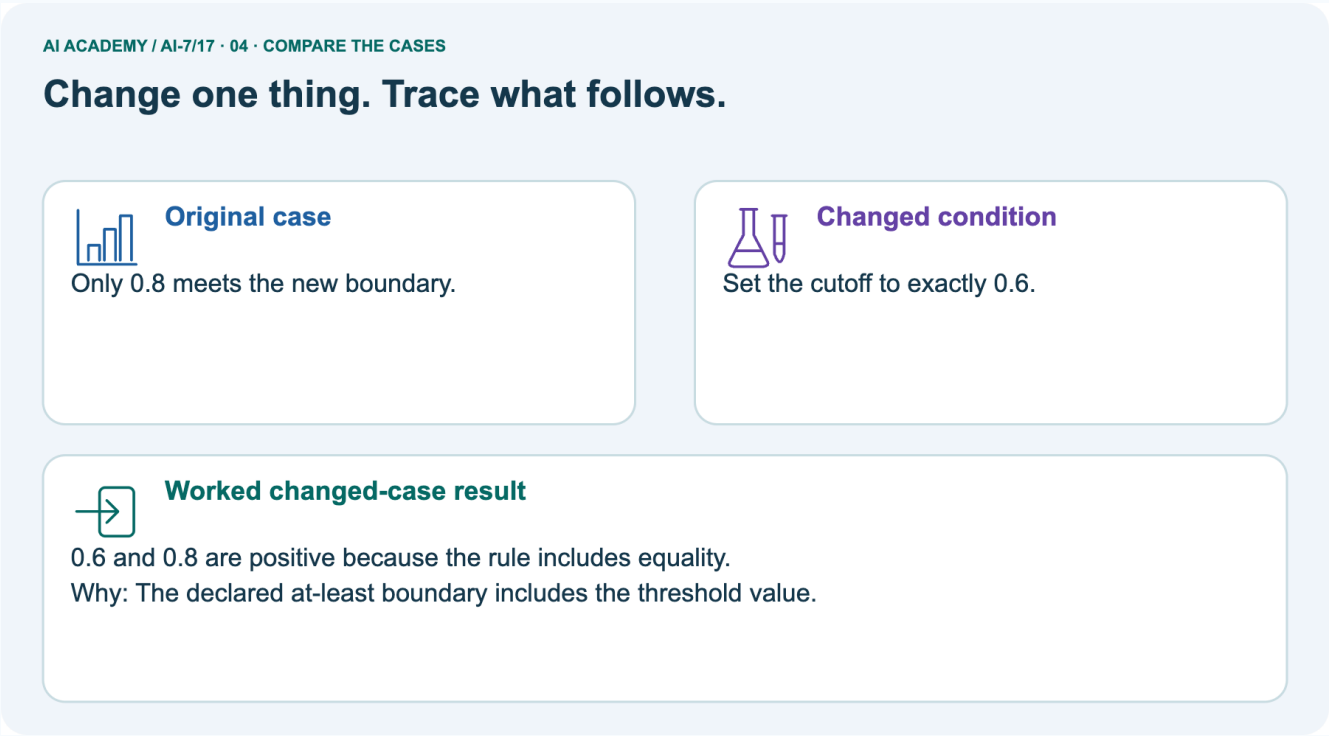

Follow the labelled picture with the matching explanation. Open the vector diagram

Score decisions under inclusive cutoffs. Cutoff 0.5: Score 0.4 = No, Score 0.6 = Yes, Score 0.8 = Yes. Cutoff 0.7: Score 0.4 = No, Score 0.6 = No, Score 0.8 = Yes. Cutoff 0.6: Score 0.4 = No, Score 0.6 = Yes, Score 0.8 = Yes.

Read the labels and values. Explain the relationship the model shows.

**My picture explanation, deciding evidence and one limit**

## 04. Change one thing. Trace what follows.




Follow the labelled picture with the matching explanation. Open the vector diagram

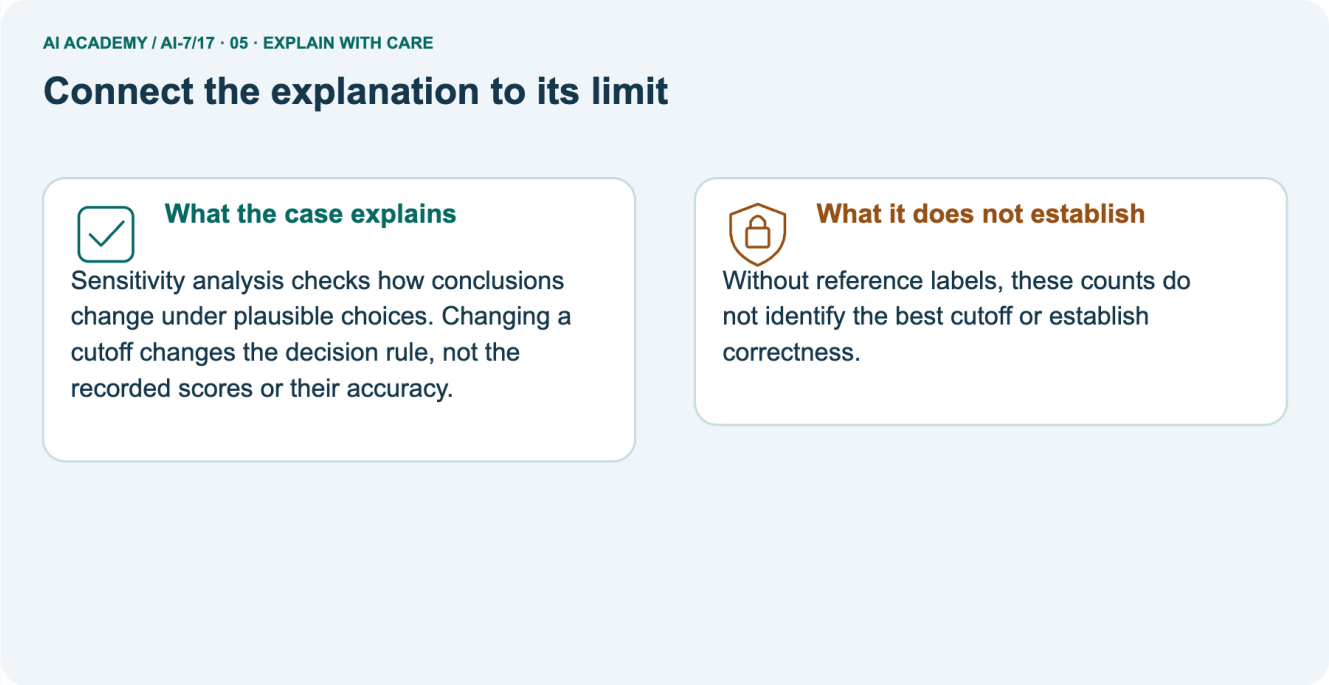
Original case: Only 0.8 meets the new boundary. Changed condition: Set the cutoff to exactly 0.6. Expected result: 0.6 and 0.8 are positive because the rule includes equality.

Reason: The declared at-least boundary includes the threshold value.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

**My picture explanation, deciding evidence and one limit**

## 05. Connect the explanation to its limit




Follow the labelled picture with the matching explanation. Open the vector diagram

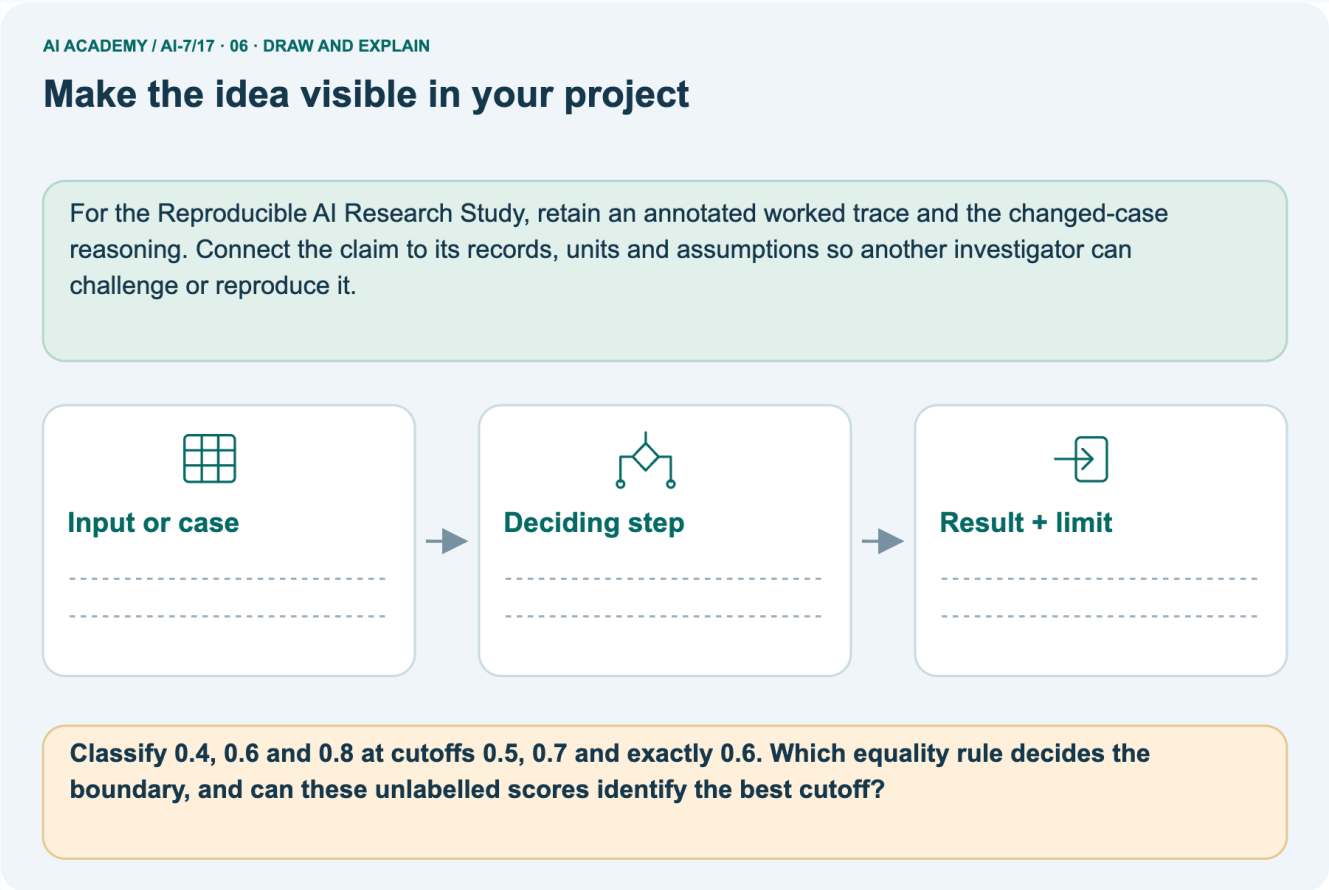
**Sensitivity analysis** checks how conclusions change under plausible choices. Changing a cutoff changes the decision rule, not the recorded scores or their accuracy.

Limit: Without reference labels, these counts do not identify the best cutoff or establish correctness.

Explain the case, then point to a limit on the claim.

**My picture explanation, deciding evidence and one limit**

## 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Classify 0.4, 0.6 and 0.8 at cutoffs 0.5, 0.7 and exactly 0.6.

Which equality rule decides the boundary, and can these unlabelled scores identify the best cutoff?

Draw your project trace and justify the deciding step using **evidence**.

**My picture explanation, deciding evidence and one limit**

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.